

Graziano Montanaro

Data Engineer

Italy · EU citizen · Open to relocation/remote

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Data engineer building distributed data platforms, production APIs, and reliable lakehouse pipelines. Open-source contributor and speaker with a research background in machine learning.

Experience

09/2024– Present **Data Engineer, Intella, Milan (Remote)**

- Set company-wide data engineering standards and built shared Python libraries used across five microservices by data, ML, and full-stack engineers, covering configuration, structured logging, storage I/O, and data quality.
- Re-architected the storage layer: migrated terabyte-scale datasets to Delta Lake and rebuilt Prefect orchestration on Kubernetes, cutting query latency up to 90% and memory use via predicate and projection pushdown in Polars.
- Delivered the API and Prefect synchronization layer for LTTB-based telemetry downsampling on TimescaleDB, enabling interactive charting over multi-year ranges.
- Built a fault-tolerant microservice orchestrating hundreds of concurrent jobs within SLA, with retries and fault isolation.

07/2026– 08/2026 **AI Tutor - Software Engineer Specialist, SpaceXAI (Freelance), Remote**

- Evaluated and critiqued Grok's software-engineering outputs against rubrics, writing reference solutions in Python.

02/2023– 09/2024 **Data Engineer, Tuidi, Putignano**

- Optimized terabyte-scale ETL with PySpark, Databricks, and Polars, cutting runtime by up to 70%.
- Built data-quality monitoring and Dash dashboards for forecasting models.

Skills

Backend	Python, SQL, Java, FastAPI, Pydantic, Apache Arrow/Flight, gRPC
Lakehouse	PySpark, Databricks, Delta Lake, Iceberg, dbt, Airflow, Prefect, Pandas, Polars, Parquet, DuckDB, DuckLake, Ibis, Great Expectations
ML & MLOps	PyTorch, TensorFlow, scikit-learn, MLflow, DVC, SHAP; time series forecasting
Streaming/DB	Kafka, RabbitMQ, PostgreSQL, TimescaleDB, MySQL, MongoDB, Redis
Cloud/DevOps	AWS (S3, Glue), Docker, Kubernetes, GitLab CI/CD, GitHub Actions, Grafana, Git
Languages	Italian (Native), English (Fluent)

Projects

Italy Dashboard, [GitHub](#)

Built a reproducible public-data platform unifying official Italian statistics from ISTAT, INPS, MUR, and weather providers.

- Implemented dbt, DuckDB, and Parquet pipelines with Python/Polars ingestion and browser-side DuckDB-WASM analytics, backed by CI-enforced provenance and frontend conformance checks.

Poor Man's Lakehouse, [GitHub](#)

Built a composable open-source lakehouse spanning local and AWS deployments.

- Unified four catalog backends, Iceberg and Delta Lake, and Spark, Polars, DuckDB, and Ibis access via a tested Python connector.

Spark Query Plan Visualizer, [GitHub](#)

- Built an interactive D3.js visualizer for Spark execution plans, surfacing optimization opportunities and bottlenecks.

Education

09/2020– 07/2022 **Master's Degree in Computer Science and Artificial Intelligence, University of Bari Aldo Moro, Bari, 110/110 cum laude (highest honours)**

Multimodal 3D CNN thesis for Alzheimer's detection from MRI and amyloid PET, reaching 95% accuracy on OASIS-3.

09/2017– 07/2020 **Bachelor's Degree in Computer Science, University of Bari Aldo Moro, Bari, 110/110 cum laude (highest honours)**

Publications

- 2024 Automated detection of Alzheimer's disease: a multi-modal approach with 3D MRI and amyloid PET. *Scientific Reports*, 14, 5210.
- 2021 Detection of Dementia through 3D convolutional neural networks based on amyloid PET. *IEEE SSCL*.